ESSENTIAL BIOLOGY FOR A'LEVEL

'THE COMPETENCY-BASED CURRICULUM'

SAMPLE BOOKLET

This booklet has been prepared to provide revision material, while also giving readers a glimpse of the quality, depth and approach of the latest full edition of *Essential Biology for A'Level: Theory & Practical*.

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- Standard practical assessment items covering biological investigations, experimental procedures, data collection, analysis and interpretation, microscopy and anatomical studies, with model responses.
- UNEB sample papers, mock examinations and seminar items, all provided with accurate model responses, giving learners exposure to different examination formats and demands.
- Self-test items for independent practice, covering both Theory and Practical Biology, to help learners assess their understanding and identify areas requiring further revision.
- Scenario-based and assessment-aligned items applying biological concepts to real-life and unfamiliar contexts, while reflecting the Elements of Construct and the demands of UNEB final papers.

The sample items that follow have been selected to give you a **glimpse of the standard of questioning and model responses** found in the latest edition.

1. Two patients presented at a busy clinic in Kampala on the same day. One was **Mukasa**, a 42-year-old non-smoking male, while the other was **Mutabaazi**, a 57-year-old male with a 12-year history of cigarette smoking, consuming at least 10 cigarettes per day. Both patients presented with shortness of breath and exercise intolerance (general body weakness).

Further examination showed that Mukasa had a hyperinflated, barrel-shaped chest but no cough, whereas Mutabaazi had a normally shaped chest but had developed a cough over the previous five years.
Genetic analysis revealed that Mukasa had a different DNA sequence in the ***SERPINA1*** gene, which is responsible for the synthesis of **α_1-antitrypsin (AAT)**.

Note: α_1-antitrypsin (AAT) is a serum protease inhibitor that normally neutralises neutrophil elastase, an enzyme released by white blood cells in the alveoli of the lungs. Mutabaazi had normal findings on genetic screening. The findings from the investigations were summarised in the table below.

A table showing the report of findings from Mukasa and Mutabaazi compared with a healthy individual

Parameter	**Mukasa (Non-smoker)**	**Mutabaazi (Smoker)**	**Healthy Individual (Non-smoker)**
Triplet code at position 342 of DNA sense strand	AAG	GAG	GAG
Blood serum AAT concentration (μg/cm³)	25.0	220.0	235.0
Relative mass of elastin tissue in the alveoli (%)	32.4	56.8	98.5
Mitochondrial oxygen consumption (nmol O_2/min/mg tissue)	41	55	100
Relative bronchial mucus production (%)	100 (normal)	285 (elevated)	100 (normal)

The attending physician advised that both patients should receive supplemental oxygen therapy through oxygen masks. However, after consultation with the senior pulmonologist, it was recommended that Mukasa should immediately begin intravenous AAT augmentation therapy, while Mutabaazi should begin antioxidant therapy and stop smoking.

Task:

(a) Critically analyse the findings in the table above and explain the different signs and symptoms experienced by:

(i) Mukasa

- ***Mukasa had AAG instead of GAG at position 342 of the DNA sense strand of the SERPINA1 gene, indicating a base substitution in which adenine replaced guanine at the first base of the triplet. This changes the codon and results in the incorporation of a different amino acid during translation, potentially altering the structure and function of α_1-antitrypsin (AAT) and reducing its ability to inhibit neutrophil elastase.***
- ***His serum AAT concentration was only 25.0 µg/cm³ compared with 235.0 µg/cm³ in the healthy individual. The low concentration indicates that insufficient functional AAT is available to neutralise neutrophil elastase released by white blood cells in the alveoli.***
- ***Consequently, neutrophil elastase remains active and progressively degrades elastin in the alveolar walls. This is reflected in Mukasa's much lower relative alveolar elastin mass of 32.4%, compared with 98.5% in the healthy individual.***
- ***The loss of alveolar elastin reduces elastic recoil of the lungs, making the alveoli less able to recoil during expiration. Air therefore becomes trapped in the lungs, resulting in hyperinflation and the barrel-shaped chest observed in Mukasa.***
- ***Mukasa's mitochondrial oxygen consumption was only 41 nmol O_2/min/mg tissue compared with 100 nmol O_2/min/mg tissue in the healthy individual. Reduced oxygen availability limits aerobic respiration and ATP production, contributing to shortness of breath and exercise intolerance/general body weakness.***
- ***His bronchial mucus production remained normal at 100%, similar to that of the healthy individual. Therefore, excessive mucus production is unlikely to be responsible for his respiratory symptoms, which is consistent with his absence of a cough.***

(ii) Mutabaazi

- ***Mutabaazi had a normal SERPINA1 gene sequence and a near-normal serum AAT concentration of 220.0 µg/cm³. This indicates that his respiratory condition was not primarily due to an inherited AAT deficiency.***
- ***His long-term cigarette smoking exposed the respiratory tissues to irritants such as tar and oxidants such as reactive oxygen species (ROS), which promote chronic inflammation and oxidative damage to lung tissue. This resulted in the reduction of relative alveolar elastin mass from 98.5% in the healthy individual to 56.8%.***
- ***Cigarette smoke also stimulated excessive mucus secretion due to irritation of the airway lining by substances such as tar, and impaired mucociliary clearance by damaging and reducing the activity of ciliated epithelial cells, which normally move mucus towards the pharynx. Consequently, mucus accumulated in the airways, as reflected by the 285% relative bronchial mucus production, compared with 100% in the healthy individual. The accumulated mucus stimulated the cough reflex in an attempt to clear it, resulting in Mutabaazi's persistent cough.***
- ***Cigarette smoke also contains carbon monoxide, which binds strongly to haemoglobin to form carboxyhaemoglobin, reducing the blood's oxygen-carrying capacity and limiting oxygen delivery to tissues. Consequently, less oxygen is available for aerobic respiration, which is***

reflected in Mutabaazi's reduced mitochondrial oxygen consumption of 55 nmol O_2/min/mg tissue, compared with 100 nmol O_2/min/mg tissue in the healthy individual. The reduced aerobic respiratory activity limits ATP production, contributing to his shortness of breath and exercise intolerance/general body weakness.

(b) Evaluate the effectiveness of the doctors' advice and recommendations for:

(i) Mukasa

- ***Supplemental oxygen therapy increases the oxygen concentration available for diffusion into the blood. This can increase oxygen delivery to body tissues and reduce breathlessness where blood oxygen levels are inadequate. However, it does not correct the AAT deficiency or prevent further destruction of alveolar elastin.***
- ***AAT augmentation therapy increases the concentration of functional AAT in the blood, allowing more neutrophil elastase to be inhibited. This reduces further degradation of alveolar elastin and can therefore slow the progression of lung tissue damage. However, it cannot restore elastin that has already been destroyed.***
- ***AAT augmentation therapy would therefore be more effective for addressing Mukasa's underlying condition, while supplemental oxygen provides supportive treatment for inadequate oxygenation.***

(ii) Mutabaazi

- ***Stopping cigarette smoking would remove the major source of respiratory irritants, oxidants and carbon monoxide, reducing further inflammation, oxidative damage and impairment of oxygen transport. This would help slow further deterioration of lung function.***
- ***Antioxidant therapy may reduce oxidative stress caused by cigarette smoke, providing some protection against further cellular and tissue damage. However, its benefit would be limited if Mutabaazi continued smoking.***
- ***Supplemental oxygen therapy could increase oxygen availability and relieve hypoxaemia where present, but it would not remove accumulated mucus or reverse the underlying smoking-induced lung damage.***
- ***Stopping smoking would therefore be the most important intervention for Mutabaazi, with antioxidant therapy providing supportive benefit. Supplemental oxygen could be used where inadequate blood oxygenation contributes to his breathlessness.***

2. A commercial vegetable farmer in Lira grows two leafy crops: a broad-leaved temperate vegetable (**Crop P**) and a tropical indigenous vegetable (**Crop Q**).
Over the past three years, the Lira region has experienced severe heat waves and prolonged drought. The farmer observed different responses among the crops. Crop P wilted severely by 1:00 PM each day, with leaves drooping completely, and its yield declined greatly within one month. Crop Q, however, maintained firm leaves throughout the day and produced a high yield despite the harsh conditions.
Concerned about these differences, the farmer collected leaf samples and sent them to an agricultural laboratory. The laboratory analysed physiological parameters of the leaves at 1:00 PM. The results are shown in the table below.

A table showing physiological parameters of crops at 1:00 PM

Physiological parameter	Crop P (Temperate)	Crop Q (Tropical Indigenous)
Endogenous abscisic acid (ABA) concentration	150 µg/g fresh weight (extremely high)	15 µg/g fresh weight (normal/low)
Stomatal pore size	Completely closed	Partially open
Rubisco affinity for oxygen	High	Very low

The physiological differences between the two crops confused the farmer and laboratory technician. The technician wants to rescue the cultivation of the high-value Crop P and proposes the following interventions:

- Installing high-pressure irrigation systems programmed to continuously spray the canopy of Crop P during the day.
- Spraying the leaves of Crop Q with a synthetic anti-transpirant designed to force complete stomatal closure during the morning hours.
- Constructing low-cost plastic greenhouse tunnels and continuously pumping artificially generated, high-purity carbon dioxide into the tunnels to increase CO_2 concentration around Crop P leaves.

Task

(a) Analyse the results and data to explain the differences in physiological activity and yield of the two crop varieties. Which crop variety should the farmer continue cultivating and why?

- ***Crop P contained an extremely high endogenous ABA concentration of 150 µg/g fresh weight, while Crop Q contained only 15 µg/g fresh weight. During drought, ABA accumulates in leaves and promotes stomatal closure to reduce water loss through transpiration. The high ABA concentration in Crop P shows severe water stress, whereas the lower ABA concentration in Crop Q suggests better regulation of water loss under the prevailing conditions.***
- ***The stomata of Crop P were completely closed, whereas those of Crop Q remained partially open. Complete stomatal closure greatly restricted carbon dioxide entry into Crop P leaves, reducing photosynthetic carbon fixation and consequently lowering carbohydrate production. In contrast, the partially open stomata of Crop Q allowed continued CO_2 uptake while still limiting excessive water loss, enabling continued photosynthesis during drought.***
- ***Crop P had a high affinity of Rubisco for oxygen, whereas Crop Q had a very low affinity for oxygen. Under the high temperatures experienced in Lira, reduced CO_2 availability around Rubisco favoured oxygen binding in Crop P. This increased the oxygenation activity of Rubisco and enhanced photorespiration, reducing carbon fixation and further lowering carbohydrate production.***
- ***In Crop Q, the low affinity of Rubisco for oxygen reduced oxygenation reactions and limited photorespiration. This allowed more efficient carbon fixation despite the high temperatures and drought conditions, contributing to continued growth and high yield.***
- ***The severe wilting of Crop P resulted from prolonged water stress, which caused high ABA accumulation and complete stomatal closure. Reduced stomatal opening limited CO_2 uptake, lowering photosynthetic carbon fixation and carbohydrate production. This reduced the***

resources available for growth and maintenance of normal leaf function, leading to the observed decline in yield.

- ***Crop Q maintained firm leaves and produced a high yield because it regulated water loss while maintaining sufficient CO_2 uptake and photosynthetic activity under drought and heat stress.***
- **Crop recommendation**: ***The farmer should continue cultivating Crop Q because it showed better adaptation to the hot and dry conditions in Lira, maintaining normal ABA levels, partially open stomata, reduced photorespiration, and high productivity. Crop P would require improved water availability and cooler conditions to achieve reliable yields.***

(b) Evaluate the practical sustainability of each intervention proposed.

- ***Continuous high-pressure irrigation of Crop P: Continuous canopy irrigation may reduce leaf temperature and water stress, lowering ABA accumulation and preventing excessive stomatal closure. However, it is expensive, requires large amounts of water, and may increase fungal diseases due to prolonged leaf wetness, making it unsustainable during drought.***
- ***Application of synthetic anti-transpirant on Crop Q: Forcing complete stomatal closure would reduce water loss, but it would also restrict CO_2 uptake and reduce photosynthetic carbon fixation. Since Crop Q is already adapted to drought through controlled stomatal opening, this intervention may reduce rather than improve productivity.***
- ***Greenhouse tunnels with continuous CO_2 enrichment for Crop P: Increasing CO_2 concentration could improve carbon fixation in Crop P by increasing CO_2 availability around Rubisco and reducing the effect of photorespiration. However, continuous CO_2 enrichment is costly, technically demanding, and does not solve the underlying water stress that causes stomatal closure.***
- ***Overall recommendation: The most sustainable approach is to promote cultivation of Crop Q under the prevailing climate while improving water management where Crop P is grown. The interventions for Crop P may improve performance temporarily but are less practical under prolonged heat and drought conditions.***

3. After attending a six-hour outdoor music festival featuring extreme noise levels and consuming large quantities of alcohol, an 18-year-old university student, David, was admitted to the university sickbay. He presented with severe disorientation, inability to walk in a straight line, profound dehydration, and complaints of a constant ringing sensation in his ears.

The university nurse assessed David and compared his physiological parameters with those of a healthy, sober student who did not attend the concert. The results are shown in the table below.

A table showing physiological parameters of David and a healthy student

Parameter	David (Post-concert and alcohol consumption)	Healthy student
ADH concentration in blood (pg/mL)	0.8	4.5
Urine volume produced over 4 hours (L)	3.5	0.6
Cardiac output (L/min)	3.2	5.1
Heart rate (beats/min)	98	75
Sensory hair cell damage in the cochlea (%)	35	0
Endolymph dynamics in the semicircular canals	Altered	Normal

Following his admission, the medical team proposed the following interventions:

- Administering isotonic saline solution intravenously.
- Giving oral rehydration salts.
- Providing plain drinking water.
- Placing David in a quiet rehabilitation environment.

The attending doctor later advised David to stop consuming alcohol to reduce the risk of long-term health complications.

Task:

(a) Analyse how David's physiological parameters differed from those of the healthy student and explain how these changes produced his clinical presentation.

- ***David had a much lower blood ADH concentration of 0.8 pg/mL compared with 4.5 pg/mL in the healthy student. Alcohol inhibited the activity of hypothalamic magnocellular neurosecretory cells, reducing the release of ADH from their axon terminals in the posterior pituitary gland. The reduced ADH concentration decreased water reabsorption in the kidney collecting ducts, causing David to produce a much larger urine volume of 3.5 L compared with 0.6 L in the healthy student. This excessive water loss resulted in profound dehydration.***
- ***The dehydration reduced blood volume, lowering venous return to the heart and decreasing cardiac output to 3.2 L/min compared with 5.1 L/min in the healthy student. David's heart rate increased to 98 beats/min compared with 75 beats/min as a compensatory mechanism to maintain blood supply to body tissues. Reduced circulation to the brain contributed to dizziness, disorientation, and impaired physical coordination.***
- ***David experienced prolonged exposure to extreme noise levels, resulting in 35% sensory hair cell damage in the cochlea, while the healthy student had no damage. Excessively loud sounds produced intense vibrations of the basilar membrane, causing overstimulation and structural damage to the delicate cochlear hair cells. Since these hair cells convert sound vibrations into electrical nerve impulses, their damage reduced accurate transmission of auditory signals to the brain. Abnormal signals from damaged hair cells were interpreted by the brain as sound even in the absence of an external stimulus, causing the persistent ringing sensation in his ears.***

- ***The endolymph dynamics in David's semicircular canals were altered, whereas those of the healthy student remained normal. Alcohol disrupted the normal movement of endolymph fluid within the semicircular canals, leading to abnormal stimulation of the sensory receptors responsible for detecting head movement. This produced inaccurate vestibular signals to the brain, impairing balance and spatial orientation, which contributed to David's disorientation and inability to walk in a straight line.***
- ***In addition, alcohol affected cerebellar function by disrupting communication between cerebellar neurons. This reduced the cerebellum's ability to coordinate skeletal muscle activity, maintain posture, and fine-tune movements, further contributing to poor coordination and unsteady walking.***
- ***Therefore, David's clinical presentation resulted from the combined effects of alcohol-induced dehydration, reduced cardiovascular performance, impaired cerebellar coordination, disruption of vestibular balance control, and noise-induced cochlear damage.***

(b) Evaluate the effectiveness of each intervention proposed by the medical team in helping David recover.

- ***Administering isotonic saline solution intravenously**: This is the most effective immediate intervention because it rapidly restores lost water and electrolytes in the bloodstream. Increasing blood volume improves venous return, cardiac output, and tissue perfusion, helping correct dehydration-related symptoms. It is safer than plain water because it maintains the normal osmotic balance of body fluids.*
- ***Giving oral rehydration salts**: Oral rehydration salts effectively replace both water and electrolytes lost through excessive urine production. Sodium and glucose promote water absorption from the intestine, improving hydration. However, it may be less effective if David is severely dehydrated or unable to drink adequately.*
- ***Providing plain drinking water**: Plain water can replace some lost fluid and improve hydration. However, it does not replace electrolytes lost through excessive urine production, and rapid intake of large amounts may disturb blood electrolyte balance. Therefore, it is less effective than isotonic saline or oral rehydration salts.*
- ***Placing David in a quiet rehabilitation environment**: A quiet environment prevents further exposure to loud sounds, reducing additional stress on the remaining cochlear hair cells and supporting recovery of temporarily affected auditory pathways. However, it cannot reverse permanent damage to sensory hair cells that have already been destroyed.*
- ***Stopping alcohol consumption**: Avoiding alcohol prevents further inhibition of ADH release, reducing excessive urine production and dehydration. It also reduces repeated impairment of cerebellar function and lowers the risk of long-term complications associated with excessive alcohol consumption.*
- ***Overall recommendation**: David should first receive intravenous isotonic saline to rapidly restore fluid and electrolyte balance, followed by oral rehydration where appropriate. A quiet environment and avoidance of alcohol are important for preventing further neurological and auditory complications, although existing cochlear hair cell damage may not be completely reversible.*

4. An illegal night fisherman was moving along the shores of the Kazinga Channel in Queen Elizabeth National Park when he stepped onto a grassy area near the water's edge. Moments later, a large territorial wild animal suddenly charged at him from a nearby thicket.

The fisherman immediately jumped backward, turned and ran at full speed across uneven ground towards a rocky slope. Visibility was very poor due to darkness, and he had to navigate the unstable terrain while moving at high speed. During the escape, he fell and sustained a deep cut on his left thigh caused by sharp rocks. Despite the injury, he continued moving for several minutes before reaching a small cave, where he remained still, lying down and experiencing severe pain. About one hour later, the rescue team arrived and took him to a nearby health post.

At the health post, the clinician injected a small volume of lignocaine around the injured area. Shortly afterwards, the affected region became completely insensitive to touch and pain, and the patient was unable to move his left leg properly for several hours. Physiological measurements were taken at different stages, and the results are shown in the table below:

A table showing physiological data of the fisherman

Evaluated parameter	Steady state (resting)	During escape	After injection
Blood flow to skeletal muscles (cm^3/min)	1,000	4,800	950
Blood flow to digestive organs (cm^3/min)	1,400	250	1,350
Membrane potential of local thigh neurones (mV)	+30	+30	-70
Sinoatrial (SA) node firing frequency (min^{-1})	70	148	82
Blood adrenaline level (ng/mL)	0.18	1.25	0.40

Task:

(a) Analyse the information above to explain how the fisherman's symptoms came about.

- ***The sudden threat and subsequent injury acted as stressors that activated the sympathetic nervous system, which stimulated the adrenal medulla to release adrenaline into the bloodstream, resulting in an increase in blood adrenaline concentration from 0.18 to 1.25 ng/mL during the escape.***
- ***The increased adrenaline stimulated the SA node, increasing its firing frequency from 70 to 148 min^{-1}, thus increasing heart rate and cardiac output. This enabled more blood to be***

delivered to the skeletal muscles, increasing the supply of oxygen and glucose for aerobic respiration, thereby supporting increased ATP production required for muscle contraction during the escape.

- ***Adrenaline also caused vasodilation of blood vessels supplying active skeletal muscles and vasoconstriction of blood vessels supplying the digestive organs. Consequently, blood flow to skeletal muscles increased from 1,000 to 4,800 cm^3/min, while blood flow to digestive organs decreased from 1,400 to 250 cm^3/min. This redistribution of blood directed a greater proportion of the available oxygen and glucose to the active muscles, supporting aerobic respiration and ATP production for sustained muscle contraction during the escape.***
- ***The deep cut caused tissue damage, stimulating pain receptors (nociceptors) in the injured thigh, which converted the pain stimulus into electrical impulses. These impulses were transmitted along sensory neurones to the spinal cord and then relayed through ascending pathways to the brain, where they were consciously perceived as severe pain. The membrane potential of the local thigh neurones reached +30 mV during the escape because depolarisation generated and propagated action potentials carrying the pain impulses.***
- ***After the injection, the membrane potential of the local thigh neurones fell to –70 mV, leaving them in their polarised resting state. This occurred because lignocaine blocked voltage-gated sodium channels, preventing sufficient sodium-ion influx to cause depolarisation and generate and propagate action potentials. Consequently, sensory impulses from the injured thigh could not effectively pass through the affected region to the spinal cord and brain, resulting in loss of sensation and pain in the affected region.***
- ***Lignocaine also blocked voltage-gated sodium channels in the motor fibres supplying the left leg, preventing the propagation of action potentials through the affected region. Consequently, action potentials travelling from the CNS along the motor neurones could not effectively reach the skeletal muscle fibres. This reduced stimulation of the muscles, impairing muscle contraction and causing the fisherman to lose proper movement of his left leg.***
- ***After the injection, the fall in adrenaline concentration from 1.25 to 0.40 ng/mL reduced stimulation of the SA node, causing its firing frequency to fall from 148 to 82 min^{-1}. Consequently, blood flow to skeletal muscles fell to 950 cm^3/min, while blood flow to digestive organs increased to 1,350 cm^3/min, as the body returned towards its resting state.***

(b) Assess the effectiveness of the injection medicine received by the fisherman and propose other interventions that could improve his chances of survival.

- ***Lignocaine was effective in providing temporary local anaesthesia because it blocked voltage-gated sodium channels in peripheral nerve fibres around the injured area. This prevented the generation and propagation of action potentials carrying pain impulses to the central nervous system, thereby reducing pain sensation. However, lignocaine did not repair the damaged tissues, control blood loss or prevent bacterial infection.***
- ***The wound should therefore be assessed, cleaned and appropriately dressed, with bleeding controlled and the wound closed where clinically appropriate. Cleaning removes microorganisms and foreign material from the wound, while appropriate wound closure promotes tissue repair and reduces the risk of bacterial infection.***
- ***The fisherman should be assessed for excessive blood loss and signs of hypovolaemic shock. If significant blood loss has occurred, appropriate intravenous fluids and, where necessary, blood transfusion would help restore circulating blood volume and blood pressure, thereby maintaining adequate blood flow and oxygen delivery to vital organs.***

- ***Tetanus prophylaxis and appropriate antimicrobial treatment should be considered according to clinical assessment. These measures would reduce the risk of tetanus and bacterial infection following the deep wound.***
- ***The fisherman should be closely monitored for complications such as continued bleeding, shock and wound infection. Early detection and treatment of these complications would improve his chances of recovery and survival.***
- ***Therefore, lignocaine was effective for temporary relief of pain but was not sufficient as the definitive treatment. Proper wound management, control of blood loss and prevention of infection would be essential for recovery and survival.***

5. John was persistently wheezing and feeling fatigued. He was then taken to a specialized hospital for medical care. Based on his signs and symptoms, the doctor suspected a condition affecting the epithelial tissue lining John's respiratory system. Normally, the cell membrane of the epithelial cells contains a protein that regulates water movement into and out of the epithelial cells. The synthesis of this protein is controlled by a gene called the **cystic fibrosis transmembrane conductance regulator (CFTR) gene**.
A defect in the CFTR gene can affect the function of the protein. Since the doctor suspected a defect in the protein, he requested a CFTR gene test and other parameters. The laboratory report from the doctor's request is shown in the table below:
A table showing John's laboratory report

Parameter	John's results	Normal range
Sequence of CFTR gene	ATG, GCT, TTT, GAC, ACA	ATG, GCT, CTT, GAC, ACA
Rate of water movement out of epithelial cells	Low	High
Mitochondrial activity	Low	High
Thickness of mucus on epithelial cells	Very thick	Thin
Status of airways	Blocked	Open

The doctor recommended that John take a coenzyme supplement for fatigue and either gene therapy or dilators for persistent wheezing. (Dilators act on the muscle cells of the bronchioles, causing them to relax and widen for some time.)
Your classmates saw John's laboratory report and the doctor's recommendations and would wish to understand what they mean in relation to human health.

Task:

(a) Analyse the laboratory report in relation to John's signs and symptoms.

- ***John's CFTR gene contains a base substitution mutation (CTT to TTT), which alters the amino acid sequence and tertiary structure of the CFTR protein, producing a non-functional protein that affects chloride ion movement across the epithelial cell membrane.***
- ***The mutated CFTR gene transcribes altered mRNA due to the substituted base sequence.***

- ***The mRNA moves from the nucleus to the rough endoplasmic reticulum, where it attaches to ribosomes for translation.***
- ***During translation, tRNA molecules bring amino acids according to the mRNA codons. Due to the altered base sequence, a different amino acid is incorporated, resulting in a protein with a different amino acid sequence.***
- ***The protein chain then folds into its three-dimensional structure. However, the altered amino acid sequence causes incorrect folding, making the protein non-functional.***
- ***The wrong protein is destroyed by enzymes within the cell, resulting in the absence of functional chloride channels in the epithelial cell membrane.***
- ***Since functional chloride channels are absent, chloride ions cannot move out of the airway epithelial cells. This prevents the formation of an osmotic gradient required for water movement into the mucus, resulting in the production of thick, dehydrated mucus that accumulates in the airways.***
- ***The thick mucus blocks and narrows the airways, causing turbulent airflow, which results in wheezing. The accumulated mucus also stimulates the cough reflex, causing a persistent cough.***
- ***Blocked airways reduce oxygen uptake, leading to reduced oxygen availability for aerobic respiration. Consequently, less ATP is produced, as indicated by the low mitochondrial activity in the laboratory report.***
- ***The reduced ATP production cannot meet the metabolic energy requirements of John's body, resulting in fatigue.***

(b) Assess the effectiveness of the doctor's recommendations.

- ***The coenzyme supplement supports aerobic respiration and ATP production, which may improve energy availability and partially reduce fatigue. However, it does not correct the CFTR mutation, restore chloride transport, or prevent thick mucus formation. Therefore, it is useful only as supportive treatment and not a long-term solution.***
- ***Gene therapy introduces a functional CFTR gene, resulting in the production of functional proteins. This restores chloride transport, allowing water to move into the mucus and thinning it, thereby addressing the root cause. However, it may cause an immune response, as the immune system may attack the viral vector delivering the gene; the gene may not reach all affected cells, and the treatment may require repeated administration. Despite these limitations, it is the most effective long-term option.***
- ***Bronchodilators relax the smooth muscles of the bronchioles, widening the airways and providing rapid relief from wheezing. However, they do not remove mucus or correct the defect, and their effects are temporary; therefore, they provide only short-term symptomatic relief.***
- ***Gene therapy should therefore be the preferred long-term treatment, with bronchodilators providing immediate relief and coenzyme supplements helping to manage fatigue. A combination of all three, with gene therapy as the foundation, will provide the greatest benefit.***

6. In the Doho Wetlands of Butaleja District, integrated rice–fish farming supports local food security. However, farmers have increasingly applied persistent organochlorine pesticides to control rice pests.
Recently, local populations of ***Oreochromis niloticus*** (Nile Tilapia) have experienced massive die-offs due to Epizootic Ulcerative Syndrome (EUS). Fisheries geneticists discovered that fast growth rate (**G**) is dominant over slow growth rate (**g**), and EUS resistance (**E**) is dominant over susceptibility (**e**).
A testcross between a heterozygous fast-growing, EUS-resistant tilapia and a slow-growing, EUS-susceptible tilapia produced the following offspring distribution:

Offspring phenotype	Number
Fast-growing, EUS resistant	415
Slow-growing, EUS susceptible	405
Fast-growing, EUS susceptible	90
Slow-growing, EUS resistant	93

To investigate the effects of persistent organochlorine pesticides on the wetland ecosystem, ecologists measured energy assimilation and pesticide residues in organisms at different trophic levels as shown in the table below.

A table showing energy flow and pesticide residues in the wetland food chain

Organism	Net energy assimilation (kJ/m²/year)	Mean organochlorine residue (mg/kg)
Phytoplankton and rice algae	85000	0.05
Zooplankton	8100	0.65
Small juvenile tilapia	760	4.80
Mature Nile Tilapia	68	38.50

Task:
(a) Analyse the genetic data and describe the inheritance pattern responsible for the observed offspring distribution. Calculate the recombination frequency between the two genes. Explain the pattern of energy flow and pesticide distribution across the organisms in the wetland ecosystem.

- ***The inheritance of growth rate and EUS resistance in Nile Tilapia is controlled by two genes. The allele G for fast growth is dominant over g for slow growth, while the allele E for EUS resistance is dominant over e for susceptibility. The testcross involved a heterozygous fast-growing, EUS-resistant tilapia (GgEe) and a slow-growing, EUS-susceptible tilapia (ggee). Since the recessive parent produces only ge gametes, the phenotypes of the offspring directly represent the types of gametes produced by the heterozygous parent.***

- ***The majority of offspring were fast-growing and EUS-resistant, with 415 individuals, and slow-growing and EUS-susceptible, with 405 individuals. These parental phenotypes occurred in higher numbers because the G and E genes are closely linked on the same chromosome. They are usually transmitted together because crossing over is less likely to separate the chromosome segments carrying them. Therefore, the heterozygous parent produced parental gametes (GE and ge) more frequently than recombinant gametes (Ge and gE), resulting in more offspring with the original allele combinations.***
- ***The smaller number of fast-growing but EUS-susceptible offspring, 90 individuals, and slow-growing but EUS-resistant offspring, 93 individuals, resulted from recombinant gametes produced during meiosis. During prophase I, homologous chromosomes pair and crossing over occurs between non-sister chromatids of homologous chromosomes at points called chiasmata. Exchange of chromosome segments can separate the original allele combinations, producing new combinations such as Ge and gE.***
- ***Since the genes are closely located on the same chromosome, crossing over between them occurs less frequently, resulting in fewer recombinant offspring than offspring with parental phenotypes.***

The total number of offspring was:
415+ 405 + 90 + 93 = 1003

The number of recombinant offspring was:
90 + 93 = 183
The recombinant frequency was:
$\frac{183}{1003}\ X\ 100 = 18.2\%$

- ***Therefore, the recombination frequency between the growth and EUS resistance genes is 18.2%. This confirms that the genes are linked because parental phenotypes occur more frequently than recombinant phenotypes.***

(b) Propose and evaluate sustainable intervention strategies for the Doho community that address both genetic improvement of tilapia stocks and ecological stability of the wetland to ensure long-term food security.

- ***Selective breeding can be used to improve tilapia stocks by selecting and breeding individuals carrying the desirable G and E alleles for fast growth and EUS resistance. Since these traits are genetically controlled, repeated selection over generations would increase the proportion of fish with improved growth performance and disease resistance, leading to more productive fish populations.***
- ***However, selection should be managed carefully because breeding from a small number of superior individuals may reduce genetic diversity. Reduced genetic variation can limit the ability of the fish population to adapt to future environmental changes or emerging diseases.***
- ***Genetic screening of breeding stock can further improve the efficiency of selection by identifying individuals carrying favourable alleles before breeding. This allows farmers to select fish with the desired genetic characteristics more accurately. However, the approach***

requires specialised equipment, technical expertise, and financial resources, which may limit its use among small-scale farmers.

- ***Reducing the use of persistent organochlorine pesticides is also necessary to restore wetland health. Farmers should adopt integrated pest management methods, including biological control, resistant crop varieties, and targeted pest control approaches. This would reduce pesticide contamination of the wetland and lower the risk of bioaccumulation and biomagnification in aquatic food chains.***
- ***Regular monitoring of pesticide residues, water quality, and fish populations would help detect environmental changes early and guide appropriate management decisions. Although monitoring requires trained personnel and resources, it would help maintain safe fish production and protect the wetland ecosystem.***
- ***Improving integrated rice-fish farming practices can also support food security because rice and fish production can occur within the same system. Fish can utilise organisms within rice fields while contributing nutrients through their waste products. However, chemical inputs must be carefully controlled because excessive pesticide use can disrupt the ecological balance of the wetland.***
- ***The most sustainable approach is an integrated strategy combining selective breeding for fast-growing, EUS-resistant tilapia, reduction of persistent pesticide use, environmentally friendly pest management, and continuous monitoring of wetland conditions. This would improve fish productivity, reduce disease outbreaks, protect biodiversity, and maintain long-term food security for the Doho community while preserving the ecological stability of the wetland ecosystem.***

7. A commercial farmer in the Doho Rice Scheme usually grows a lowland rice variety in the paddies. Hoping to maximize carbohydrate yield for animal feed, the farmer attempts to intercrop the rice with a high-yielding maize variety.
One month after planting, heavy rains cause the fields to remain deeply waterlogged for three weeks. The farmer observes that the rice thrives and continues to grow rapidly, while the maize becomes severely stunted, its leaves turn yellow, and the root system begins to rot.
Plant physiologists analysed the structures and physiological responses of the lowland rice and maize to investigate the cause of the difference in survival. The results are summarised in the table below.

A table showing physiological and anatomical characteristics of lowland rice and maize under waterlogged conditions

Parameter	Lowland Rice	Maize
Root tissue ethylene concentration (nl/g/hr)	45.2	2.1
Root aerenchyma (air space) volume (%)	40	5
Soil oxygen concentration around roots (%)	Reduced	Reduced
Bundle sheath PEP carboxylase activity	0	15% of normal

Following the observations, agricultural officers proposed the following interventions:

- ❖ Digging shallow drainage channels in the maize fields.
- ❖ Using timed planting to target the non-rainy seasons.
- ❖ Applying moderate artificial nitrogen fertilizers.

Task:

(a) Analyse the physiological and anatomical data to explain why maize failed in waterlogged soil while rice thrived. Relate maize's failure to thrive to the observed bundle sheath PEP carboxylase activity.

- ***Waterlogging creates oxygen-deficient conditions around plant roots because water fills the spaces between soil particles and reduces the movement of oxygen into the soil. Although both lowland rice and maize experienced reduced soil oxygen concentration, rice survived because it possesses structural and physiological adaptations that allow it to maintain root respiration under flooded conditions.***
- ***Lowland rice had a higher root tissue ethylene concentration of 45.2 nl/g/hr compared with 2.1 nl/g/hr in maize. The accumulation of ethylene in rice roots stimulated the breakdown of some root cells, creating air spaces within the root tissues known as aerenchyma. These air spaces allowed oxygen to move from aerial parts of the plant to the roots. This was supported by the higher root aerenchyma volume in rice, 40%, compared with only 5% in maize. Therefore, rice maintained a supply of oxygen to root cells, allowing aerobic respiration, energy production, and continued growth despite waterlogged conditions.***
- ***In contrast, maize had limited aerenchyma formation because of its low root ethylene concentration. As a result, oxygen supply to the roots was insufficient, causing reduced aerobic respiration and lower ATP production. This affected active transport processes, mineral uptake, and root growth. Prolonged oxygen deficiency also promoted root tissue damage and rotting, which contributed to yellowing of leaves and severe stunting.***
- ***The failure of maize was also related to the reduced PEP carboxylase activity observed in the plant. Maize is a C4 plant that normally uses PEP carboxylase during the initial fixation of carbon dioxide in mesophyll cells. The enzyme catalyses the combination of carbon dioxide with phosphoenolpyruvate, forming oxaloacetate, which is converted to malate and transported to bundle sheath cells where it releases carbon dioxide for use in the Calvin cycle to produce carbohydrates. The reduction of PEP carboxylase activity to 15% of normal reduced carbon fixation, hence reducing carbohydrate production and limiting growth.***
- ***Therefore, while rice survived because its ethylene response promoted aerenchyma formation and oxygen transport to roots, maize lacked sufficient adaptations to waterlogged conditions. This resulted in impaired root respiration, reduced nutrient uptake, decreased carbon fixation, and poor growth.***

(b) Evaluate how the proposed interventions could improve crop survival and productivity in the Doho Rice Scheme.

- ***Digging shallow drainage channels in maize fields would improve crop survival by removing excess water and increasing oxygen availability in the soil. Improved soil aeration would enhance root respiration, nutrient uptake, and root growth, reducing root damage and improving maize productivity. However, drainage should be managed carefully because***

excessive removal of water may reduce water availability required by rice grown in flooded conditions.

- ***Using timed planting to target the non-rainy seasons would reduce exposure of maize to prolonged waterlogging. Planting maize when rainfall is lower would allow roots to develop under well-aerated soil conditions, improving plant establishment, growth, and yield. However, this approach depends on reliable prediction of rainfall patterns, and unexpected heavy rains may still cause flooding.***
- ***Applying moderate artificial nitrogen fertilizers could improve crop growth because nitrogen is required for chlorophyll formation, enzyme synthesis, and production of proteins involved in plant development. This may increase recovery and biomass production after stress. However, fertilizer application alone cannot overcome oxygen deficiency caused by waterlogging, and excessive use may cause nutrient runoff and pollution of surrounding water bodies.***
- ***The most effective approach would be an integrated strategy combining proper water management through drainage, suitable planting schedules, and balanced fertilizer application. This would reduce waterlogging stress, improve nutrient availability, and increase crop survival and productivity in the Doho Rice Scheme.***

8. Alex, an 8-year-old, had been persistently experiencing severe fatigue, poor blood circulation and poor physical growth compared with his peers. He was taken to a specialised hospital for comprehensive medical care. Based on his signs and symptoms, the doctor suspected a genetic condition affecting the structure of haemoglobin inside the red blood cells, which subsequently affected his normal body functioning. Since the doctor suspected a defect in haemoglobin, she requested a gene test, cellular structure evaluation and assessment of metabolic parameters.
The laboratory report from the doctor's request is shown in the table below.

A table showing Alex's laboratory report:

Parameter	Alex's results	Normal range
Sequence of haemoglobin gene segment	GGA, CAC, CTC	GGA, CTC, CTC
Shape of red blood cell	Sickle shaped	Biconcave shaped
Rate of ATP production	Low	High

The doctor recommended any one of the following interventions to address Alex's health condition:

- ❖ Taking daily iron supplements.
- ❖ A bone marrow transplant from a compatible healthy donor.
- ❖ Blood transfusions from compatible healthy donors.

Your classmates saw Alex's laboratory report and the doctor's recommendations and would like to understand what they mean in relation to human health.

Task:
(a) Analyse the laboratory report in relation to Alex's signs and symptoms.

- ***Alex's haemoglobin gene segment contains a base substitution mutation in which an adenine (A) base has replaced a thymine (T) base, changing the codon from CTC to CAC. This mutation alters the genetic information used to produce haemoglobin, resulting in abnormal haemoglobin associated with the sickle-shaped red blood cells observed in Alex.***
- ***The mutated haemoglobin gene is transcribed into altered mRNA, which moves from the nucleus to the rough endoplasmic reticulum, where it attaches to ribosomes for translation.***
- ***During translation, tRNA molecules bring amino acids according to the mRNA codons. The altered codon causes a different amino acid to be incorporated into the haemoglobin polypeptide, changing its amino acid sequence.***
- ***The altered polypeptide folds into an abnormal three-dimensional structure, producing haemoglobin with altered properties such as having the s shape, which causes red blood cells to become sickle-shaped rather than normally biconcave.***
- ***The sickle-shaped red blood cells are less flexible and can obstruct narrow blood vessels, reducing blood flow to tissues. This accounts for Alex's poor blood circulation and reduces the delivery of oxygen required by body cells for aerobic respiration.***
- ***Reduced oxygen delivery limits aerobic respiration and ATP production, as indicated by Alex's low rate of ATP production. The reduced ATP availability contributes to severe fatigue and limits the energy available for normal cellular activities, including growth and tissue development, contributing to his poor physical growth.***

(b) Assess the effectiveness of the doctor's recommendations.

- ***<u>Taking daily iron supplements</u> could support haemoglobin synthesis if Alex had inadequate iron intake because iron is an essential component of haemoglobin. However, supplementation would not correct the mutation in the haemoglobin gene or restore production of normal haemoglobin. Sickle-shaped red blood cells would therefore continue to form, making iron supplementation insufficient to address the underlying condition.***
- ***<u>A bone marrow transplant</u> from a compatible healthy donor would be the most effective option for addressing the underlying genetic problem. The transplant can replace Alex's abnormal blood-forming stem cells with healthy stem cells that produce red blood cells containing normal haemoglobin, thereby reducing the formation of sickle-shaped cells and improving oxygen transport. However, it requires a suitably matched donor and may cause complications such as rejection of the transplanted cells, infection due to temporary weakening of the immune system, or damage to body tissues.***
- ***<u>Blood transfusions</u> from compatible healthy donors would temporarily increase the number of normal red blood cells containing functional haemoglobin. This would improve oxygen transport to tissues and help reduce fatigue, but it would not correct the mutated haemoglobin gene, so abnormal cells would continue to be produced.***
- ***Therefore, a compatible bone marrow transplant would be the most effective of the three options because it addresses the underlying genetic problem, while blood transfusions would mainly provide temporary relief by improving oxygen transport.***

9. A coach of a certain sports team wanted to improve the performance of his athletes during sports. To do this, he asked a physiologist to provide him with essential information on how blood supply to different body regions changes during exercise so that he could design special exercises and give better advice to his athletes.
The data provided by the physiologist are shown in **the table** below.

A table showing changes in blood supply and heart rate at rest and during exercise:

Parameter	At rest	During exercise
Blood supply to skeletal muscles (cm^3/minute)	1,200	12,500
Blood supply to skin (cm^3/minute)	500	1,900
Blood supply to brain (cm^3/minute)	750	750
Heart rate (beats per minute)	72	100

The physiologist later suggested any one of the following strategies to be adopted by the coach to improve the performance of his athletes:

- ❖ Daily training at high altitude and eating glucose solutions a few minutes before exercise.
- ❖ Daily training at sea level and eating a heavy carbohydrate meal a few minutes before exercise.

Task:

(a) Analyse the physiological changes in the athlete's body during exercise.

- ***Blood supply to skeletal muscles increased from 1,200 cm^3/minute at rest to 12,500 cm^3/minute during exercise. This increased blood flow delivers more oxygen and glucose to the muscles for increased aerobic respiration and ATP production, while facilitating the removal of carbon dioxide and other metabolic wastes produced during vigorous muscle contraction.***
- ***Blood supply to the skin increased from 500 cm^3/minute at rest to 1,900 cm^3/minute during exercise. This increase results from increased heat production by actively respiring muscles, which raises body temperature and triggers cutaneous vasodilation. More warm blood is therefore directed to the skin, enhancing heat transfer to the body surface and facilitating heat loss to the surroundings, thereby limiting the increase in core body temperature.***
- ***Blood supply to the brain remained constant at 750 cm^3/minute at rest and during exercise. This is because cerebral blood flow is tightly regulated by autoregulation to maintain a relatively constant supply of oxygen and glucose to brain cells despite changes in the body's overall blood-flow demands.***
- ***Heart rate increased from 72 to 100 beats per minute during exercise. The increased heart rate raises cardiac output, increasing the volume of blood delivered to the skeletal muscles and skin per minute to meet their increased physiological demands.***

(b) Evaluate the effectiveness of each strategy given by the physiologist and advise the coach.

- ***Daily training at high altitude would stimulate increased production of red blood cells and haemoglobin, increasing the oxygen-carrying capacity of the blood and improving oxygen delivery to skeletal muscles for aerobic respiration. However, consuming glucose solution only a few minutes before exercise would provide readily available glucose but would not provide sufficient energy for prolonged exercise.***
- ***Daily training at sea level would provide a higher partial pressure of oxygen but would not stimulate the same increase in red blood cell and haemoglobin production induced by high-altitude training. Eating a heavy carbohydrate meal only a few minutes before exercise would also provide limited immediate benefit because digestion and absorption would not occur sufficiently quickly, and the large meal could cause gastrointestinal discomfort during exercise.***
- ***Therefore, the coach should adopt the high-altitude training strategy, while providing an appropriate carbohydrate source before exercise rather than relying on a heavy meal immediately before exercise. This would increase the oxygen-carrying capacity of the blood and provide glucose for ATP production during exercise.***

10. Last month, Okello and Musa were involved in a road accident. Okello was admitted to a nearby clinic, while Musa was taken to a referral hospital. Each patient was managed as shown below.

- **Okello** was administered all recommended vaccines one week before surgery and was given morphine 48 hours before surgery.
- **Musa** was not vaccinated before surgery and was not given a painkiller during surgery because of an allergy.

Morphine is a painkiller that binds to receptors on the presynaptic membrane of a cholinergic synapse and prevents the entry of calcium ions into the presynaptic knob. The patients' conditions during and after surgery are shown in the table below.

Table: Patients' conditions during and after surgery

Parameter	Okello	Musa
Pain perception during surgery	Less	Much
Condition after 10 days	Healing was successful	No healing yet; secondary infections at the site of surgery and fever
Cardiac output at day 10 (dm^3 per minute)	5.2	6.8

Some medical interns later advised that, to ensure faster recovery for Musa, one of the following strategies could have been used:

- Administer Musa with lidocaine, a regional nerve block that blocks sodium ion channels in nerve axons, instead of morphine, and administer vaccines with specific antibodies on the day of surgery.

- ❖ Give Musa an oral mild painkiller one hour before surgery and administer antibacterial drugs after visible signs of inflammation.

Task:

(a) Analyse the patients' conditions during and after surgery as shown in the table above in relation to how they were managed by the different health facilities.

- ***Okello experienced less pain during surgery because morphine bound to receptors on the presynaptic membrane of a cholinergic synapse and reduced the entry of calcium ions into the presynaptic knob. This reduced the release of neurotransmitter into the synaptic cleft, so the pain impulse was not transmitted across the synapse.***
- ***Musa experienced much more pain during surgery because he was not given a painkiller due to his allergy. Pain impulses were therefore transmitted through sensory neurones to the central nervous system, where they were interpreted, resulting in greater pain perception.***
- ***Okello had healed successfully after 10 days, whereas Musa had not healed and had developed secondary infections at the surgical site and fever. Vaccination one week before surgery had allowed Okello to develop specific antibodies against the relevant pathogens, enabling a faster immune response and reducing the risk of infection. Musa lacked this prior protection, making infection more likely. The resulting infection and fever increased metabolic demands, diverting oxygen, nutrients and energy towards immune defence and temperature regulation, thereby delaying tissue repair.***
- ***Musa had a higher cardiac output of 6.8 dm^3/minute compared with 5.2 dm^3/minute in Okello at day 10. The higher cardiac output supplied more oxygen and nutrients to tissues to meet the increased metabolic demands of infection and fever, while increased blood flow to the skin facilitated heat loss so as to prevent excessive rise in body temperature.***

(b) Evaluate the strategies given by the medical interns and choose the one that would have been most effective in bringing about faster recovery in Musa.

- ***Administering lidocaine instead of morphine would provide effective pain control because lidocaine blocks sodium ion channels in sensory neurones, preventing the generation and transmission of action potentials and therefore reducing pain transmission during surgery. Administering specific antibodies would provide immediate passive immunity against the relevant pathogens, reducing the risk of infection. However, vaccination on the day of surgery would not allow sufficient time for active immunity to develop.***
- ***Giving Musa an oral mild painkiller one hour before surgery would reduce pain perception, but may not provide adequate pain control throughout the operation. Administering antibacterial drugs after visible signs of inflammation would allow the infection to become established before treatment, giving pathogens time to multiply and potentially delaying wound healing.***
- ***Therefore, administering lidocaine together with specific antibodies would have been more effective in promoting faster recovery because it would provide effective pain control and immediate protection against infection, reducing factors that could delay tissue repair.***

11. The diamondback moth, *Plutella xylostella*, is a major pest of cabbage crops in tropical agricultural regions. Rising temperatures and the widespread use of synthetic pesticides have increased the challenge of controlling this pest. Scientists investigated the

inheritance of pesticide resistance and wing morphology under different environmental conditions.
Two inherited genes were studied:

- **Locus R/r:** Pesticide resistance (**R**-resistant, **r**-susceptible)
- **Locus W/w:** Wing morphology (**W**-long wings, **w**-wrinkled wings)

A true-breeding resistant, long-winged moth (**RRWW**) was crossed with a true-breeding susceptible, wrinkled-winged moth (**rrww**). All F_1 offspring were resistant and long-winged, with genotype **RrWw**.

To investigate whether environmental temperature influences the transmission of these genes, adult F_1 moths were divided into two groups and acclimatised to different temperatures before each group was test-crossed with homozygous recessive moths (**rrww**).

- **Condition A:** Test cross conducted at **22°C**, the baseline temperature.
- **Condition B:** Test cross conducted at **32°C**, representing prolonged exposure to higher temperatures associated with global warming.

A table showing Phenotypic distribution of test-cross offspring

Phenotype	Condition A (22°C)	Condition B (32°C)
Resistant, long-winged	470	500
Susceptible, wrinkled-winged	468	500
Resistant, wrinkled-winged	32	0
Susceptible, long-winged	30	0

Over a 10-year period, farmers repeatedly sprayed a synthetic chemical pesticide. Ecologists monitored the frequency of the resistant allele **R** in the moth population and the population size of a native predatory beetle that naturally controls the moth population.

A table showing ten-year trends in resistance, predator population and ecosystem health

Year	Frequency of resistant allele (R)	Predator population (per km²)
1	0.05	1,200
4	0.22	650
7	0.58	110
10	0.95	0

Ecologists have proposed the following strategies to restore the ecosystem:

- Apply highly selective pesticides.
- Introduce and conserve predatory beetles.
- Practise crop rotation.

Task:

(a) Analyse how increased temperature and continuous pesticide application have influenced the transmission patterns of the moth's genes, the frequency of the resistant allele and the population of predatory beetles.

- ***The test cross involved an F_1 moth with genotype RrWw and a homozygous recessive moth with genotype rrww. Since the recessive parent could produce only rw gametes, the phenotypes of the offspring reflected the types of gametes produced by the F_1 moth.***
- ***At 22°C, 470 offspring were resistant with long wings and 468 were susceptible with wrinkled wings. These were the parental phenotypes because they resulted from the parental gametes RW and rw, which retained the original allele combinations. The smaller numbers of resistant, wrinkled-winged offspring (32) and susceptible, long-winged offspring (30) resulted from recombinant gametes, Rw and rW, produced when crossing over occurred between the two loci during meiosis.***
- ***At 32°C, 500 offspring were resistant with long wings and 500 were susceptible with wrinkled wings, while no recombinant phenotypes were produced. This indicates that crossing over between the R/r and W/w loci did not occur under the conditions tested. Consequently, the F_1 moth transmitted the parental gamete types RW and rw, resulting in only the parental phenotypes among the offspring.***
- ***Continuous pesticide application increased the frequency of the resistant R allele from 0.05 in year 1 to 0.22 in year 4, 0.58 in year 7 and 0.95 in year 10. The pesticide acted as a selection pressure, causing greater mortality among susceptible moths. Resistant moths were more likely to survive and reproduce, so they contributed more R alleles to subsequent generations, increasing the frequency of R in the population.***
- ***The predatory beetle population declined from 1,200 individuals per km^2 in year 1 to 650 in year 4, 110 in year 7 and none by year 10. Continuous pesticide application likely reduced the survival and reproduction of these non-target organisms through their exposure to the pesticide, weakening natural predation of the diamondback moth. Consequently, the decline in beetle numbers reduced biological control of the pest population.***

(b) Evaluate the effectiveness of each proposed strategy in promoting long-term crop productivity.

- ***Applying highly selective pesticides would reduce damage to predatory beetles and other non-target organisms compared with non-selective pesticides. This would help conserve natural predators and maintain biological control of the diamondback moth. However, repeated pesticide use would continue to impose selection pressure on the moth population, favouring resistant individuals and potentially increasing the frequency of the R allele. Therefore, selective pesticides would be useful but should not be relied upon as the sole control measure.***
- ***Introducing and conserving predatory beetles would strengthen biological control by increasing predation on diamondback moths and reducing their population without directly selecting for pesticide resistance. However, the beetles would need protection from pesticide exposure and suitable conditions for their survival and reproduction.***
- ***Practising crop rotation would reduce the continuous availability of cabbage as a host plant, disrupting the feeding and breeding of the diamondback moth and reducing its population between cropping seasons. It would also reduce continuous pesticide-selection pressure,***

helping to slow the further increase of resistance. However, its effectiveness would depend on the crops used in the rotation and whether the moth can feed and reproduce on those crops.

- ***A combination of introducing and conserving predatory beetles with crop rotation and limited use of highly selective pesticides would provide the most sustainable approach. It would strengthen natural biological control, disrupt the moth's life cycle and reduce selection pressure for pesticide resistance, thereby supporting long-term crop productivity.***

12. Lake Victoria is shared by Uganda, Kenya, and Tanzania. It was once famous for its remarkable biodiversity, supporting over 500 unique species of cichlid fish that evolved from a common ancestor through adaptive radiation. In the 1950s, the predatory Nile perch (*Lates niloticus*) was introduced into the lake. Soon after, the invasive water hyacinth (*Eichhornia crassipes*) began spreading rapidly across the water surface. These changes altered the lake environment, creating strong selective pressures on native fish populations. Ecologists and geneticists monitored the lake over several decades to investigate how these pressures affected both the ecosystem and the genetic traits of surviving cichlid populations.
The key data collected are presented in the table below.

A table showing changes in Nile perch population, water hyacinth cover and cichlid populations

Year	Nile perch population	Water hyacinth cover (%)	Number of active cichlid species	Frequency of G-gene
1955 (before introduction)	0	0	500+	0.05
1960	4,000	0	380	0.06
1980	200,000	5	200	0.28
2026	350,000	15	150	0.82

The **G-gene** controls a larger gill surface area in cichlid fish.

The conservation team has proposed two strategies to restore the cichlid population and genetic diversity and is uncertain about which would be more effective.

- Capture a representative proportion of the remaining cichlid species and breed them in a controlled, highly oxygenated and Nile perch-free environment. Through selective breeding, further increase the frequency of the **G-gene**, apply herbicides to control water hyacinth, and remove as much Nile perch from the lake as possible.
- Capture a representative proportion of the remaining cichlid species and breed them in a controlled, highly oxygenated and Nile perch-free environment. Through selective breeding, increase the frequency of the ancestral **g-gene**,

introduce specific biological control agents to reduce water hyacinth, and regulate the Nile perch population.

Task:

(a) (i) Explain how changes in Nile perch population and water hyacinth cover have affected the number of active cichlid species.

- ***Before the introduction of Nile perch in 1955, Lake Victoria supported over 500 active cichlid species, with no water hyacinth cover. The absence of the introduced predator allowed the cichlid populations to occupy different ecological niches with relatively little predation pressure.***
- ***By 1960, the Nile perch population had increased to 4,000, while water hyacinth cover remained at 0%. During the same period, the number of active cichlid species decreased from over 500 to 380. The increase in Nile perch introduced predation pressure on native cichlids, particularly smaller and vulnerable species, reducing their survival and population sizes and contributing to the loss of some species.***
- ***By 1980, the Nile perch population had increased greatly to 200,000, while water hyacinth covered 5% of the lake surface. The number of active cichlid species fell further to 200. Increased predation reduced cichlid populations, while water hyacinth formed dense mats that reduced light penetration and restricted gas exchange at the water surface. This reduced photosynthesis by aquatic producers and contributed to lower dissolved oxygen concentrations, creating additional environmental stress for cichlids.***
- ***By 2026, the Nile perch population had reached 350,000 and water hyacinth cover had increased to 15%, while only 150 active cichlid species remained. The continued increase in predation pressure, together with habitat deterioration and reduced oxygen availability associated with expanding water hyacinth, further reduced the survival and distribution of native cichlids.***

(ii) Explain how increased water hyacinth cover has affected the frequency of the **G-gene** in the cichlid fish.

- ***The frequency of the G-gene increased from 0.05 in 1955 to 0.06 in 1960, 0.28 in 1980 and 0.82 in 2026 as water hyacinth cover increased from 0% to 15%.***
- ***Increasing water hyacinth cover reduced light penetration into the water and contributed to lower dissolved oxygen levels by restricting gas exchange and through decomposition of accumulated plant material. This created hypoxic conditions that placed greater respiratory stress on cichlid fish.***
- ***Cichlids carrying the G-gene had a larger gill surface area, providing a greater surface through which oxygen could diffuse into the blood. Under conditions of reduced dissolved oxygen, these fish were therefore better able to obtain oxygen and maintain aerobic respiration than fish with smaller gill surfaces.***
- ***Consequently, cichlids carrying the G-gene had a survival advantage in the increasingly oxygen-limited environment and were more likely to reproduce and pass the G-gene to their offspring. Natural selection therefore increased its frequency from 0.05 to 0.82.***

(b) Evaluate the effectiveness of each strategy proposed by the conservation team, select the more sustainable strategy and justify your choice.

- ***Controlled breeding of representative cichlid populations in a highly oxygenated, Nile perch-free environment would protect the fish from predation and provide sufficient oxygen for aerobic respiration, thereby improving their survival, growth and reproduction.***
- ***Selective breeding for the G-gene would increase the frequency of the allele associated with a larger gill surface area, potentially improving oxygen uptake under low-dissolved-oxygen conditions. Removing Nile perch would further reduce predation pressure and allow more cichlids to survive and reproduce.***
- ***However, extensive removal of Nile perch may be difficult to sustain because the fish reproduce and spread readily in the lake. Herbicides used to control water hyacinth could also harm non-target aquatic organisms because they may affect organisms other than the weed, while their residues could introduce chemical pollutants into the lake.***
- ***Controlled breeding combined with selective breeding for the ancestral g-gene, biological control of water hyacinth and regulation of Nile perch would also reduce the major pressures on cichlids. However, increasing the frequency of the g-gene is less suitable because the G-gene is associated with a larger gill surface area and may favour survival under the low-dissolved-oxygen conditions associated with increasing water hyacinth cover.***
- ***Biological control of the weed and regulation of Nile perch would be less disruptive to the lake ecosystem than herbicide application and extensive predator removal.***
- ***Therefore, controlled breeding combined with selective breeding for the G-gene, reduction of water hyacinth and removal of Nile perch would be the more effective strategy for restoring the cichlid population and its genetic traits.***

13. Due to frequent changes in climatic conditions that lead to poor crop yields, farmers have been advised to start using greenhouses for tomato cultivation. A greenhouse is a house-like structure covered with transparent or translucent walls and a roof that create a controlled environment for growing crops.
To determine the most suitable conditions for use in greenhouses, the farmers hired an agricultural officer, who conducted a trial on tomato growing using four greenhouses. Each greenhouse was maintained under different conditions, and the results of the trial are shown in the table below:

A table showing tomato growth under different greenhouse conditions

Growth and physiological traits	Greenhouse 1 (32°C + auxins)	Greenhouse 2 (32°C + gibberellins)	Greenhouse 3 (25°C + auxins)	Greenhouse 4 (25°C + gibberellins)
Germination rate (%)	Moderate	Moderate	Moderate	High
Photosynthetic rate (µmol $CO_2/m^2/s$)	High	High	Low	Low
Number of branches	Few	Very many	Few	Many

For good yields, the agricultural officer recommended that the greenhouse conditions should not be kept constant throughout the growing period. The farmers need to know how to implement this recommendation.

Task:

(a) Analyse the relationship between the greenhouse conditions and the growth and physiological traits of tomatoes.

- ***Germination is highest in Greenhouse 4, maintained at 25°C with gibberellins, because germination enzymes work close to their optimum temperature at 25°C, whereas at 32°C, enzyme activity is reduced due to partial denaturation.***
- ***In Greenhouse 4, gibberellins stimulate the production of amylase, which hydrolyses stored starch into sugars for respiration and embryo growth. Therefore, the combination of optimum temperature and gibberellins results in the highest germination rate.***
- ***The photosynthetic rate is high in the greenhouses maintained at 32°C and low in those maintained at 25°C, regardless of the growth regulator used, because photosynthetic enzymes such as RuBisCO are more active at 32°C, increasing carbon fixation and glucose production.***
- ***Therefore, temperature is the main factor influencing the rate of photosynthesis, while the growth regulators have little influence on photosynthesis.***
- ***Plants treated with gibberellins produce more branches, with the highest number recorded in Greenhouse 2, maintained at 32°C with gibberellins, because gibberellins promote lateral bud growth, whereas auxins maintain apical dominance.***
- ***The higher temperature also increases the rate of enzyme-controlled growth processes, resulting in the greatest number of branches in Greenhouse 2.***

(b) Justify the recommendation of the agricultural officer and advise the farmers.

- ***No single greenhouse condition is best for all three parameters: germination rate, photosynthetic rate, and number of branches. Therefore, different conditions are suitable for different growth stages.***
- ***Greenhouse 4, maintained at 25°C with gibberellins, gives the highest germination rate, producing the greatest number of seedlings from seeds. However, because the photosynthetic rate is low at 25°C, it is unsuitable for later growth and fruiting. Therefore, it is best used during the germination stage only.***
- ***Greenhouse 2, maintained at 32°C with gibberellins, gives the highest photosynthetic rate and the greatest number of branches, providing more sites for fruit production. However, its lower germination rate makes it more suitable after seedlings have been established.***
- ***Therefore, farmers should use the conditions of Greenhouse 4 during germination and then switch to the conditions of Greenhouse 2 during growth and branching. Gibberellins should be maintained throughout, while auxins should not be used as the main treatment because they suppress branching and do not improve germination.***

14. Aisha, a young adult managing Type 1 Diabetes mellitus, attends regular medical check-ups, including receiving vaccinations at a local health centre.
Following a recent vaccination against hepatitis B that she received, her blood showed a successful immune response: her antibody concentration increased from 50 units per 100 ml to 275 units per 100 ml of blood.
Since Type 1 Diabetes management involves dealing with metabolic and immune challenges, her doctor also reviewed the effects of body temperature on oxygen-carrying capacity and antibody production.

A table showing effects of body temperature on oxygen-carrying capacity and antibody production

Body temperature (°C)	Oxygen-carrying capacity of blood (ml of O_2/100 ml of blood)	Antibody production
38	0.38	High
37	0.40	Moderate
35	0.46	Low

The medical doctor is suggesting any one of the following interventions to address Aisha's health condition:
i. Aisha eliminates carbohydrates from her diet.
ii. Aisha takes insulin injections 30 minutes after every carbohydrate-containing meal.
iii. Aisha takes a lipid-rich diet.
Aisha needs to understand the effectiveness of each intervention before making a choice.

Task:
(a) Analyse how vaccination and changes in body temperature affected the functioning of Aisha's body.

- ***Antibody concentration rose from 50 to 275 units per 100 ml because vaccine antigens activated specific B lymphocytes, which underwent clonal expansion and differentiated into plasma cells that produced antibodies. Some B lymphocytes became memory cells; therefore, the rise shows a successful immune response.***
- ***Oxygen-carrying capacity decreased as temperature rose because haemoglobin's affinity for oxygen decreases at higher temperatures, causing more oxygen to be released to tissues. Therefore, during fever, less oxygen is carried in the blood.***
- ***Antibody production increased with temperature, being highest at 38°C, because antibody production involves enzyme-controlled processes, and a mild increase in temperature can increase the rate of immune reactions.***
- ***However, very high temperatures may denature enzymes and impair antibody synthesis and other cellular processes. Therefore, although a mild increase in temperature may enhance antibody production, it can reduce the oxygen-carrying capacity of blood, limiting oxygen supply to tissues.***

(b) Evaluate the effectiveness of each intervention suggested by the doctor in the management of Aisha's condition.

- ***Aisha has Type 1 Diabetes mellitus, in which pancreatic β-cells produce little or no insulin. Therefore, effective treatment should replace the missing insulin and maintain normal blood glucose levels.***
- ***Eliminating carbohydrates from her diet reduces the amount of glucose absorbed into the blood after meals, but it does not replace the missing insulin or restore normal glucose metabolism. As a result, the body relies more on fat metabolism for energy, increasing the production of ketone bodies, which can lower blood pH and lead to diabetic ketoacidosis. Therefore, it is not an effective treatment for Type 1 Diabetes mellitus.***
- ***Insulin injections replace the missing hormone, promoting glucose uptake by body cells and the conversion of excess glucose into glycogen in the liver and muscles. This lowers blood glucose and directly addresses the underlying cause of Type 1 Diabetes. However, taking insulin 30 minutes after every carbohydrate-containing meal may delay blood glucose control, while an incorrect dose may cause hyperglycaemia or hypoglycaemia. When correctly timed and matched with carbohydrate intake, insulin therapy is the most effective intervention.***
- ***A lipid-rich diet reduces carbohydrate intake but does not replace the missing insulin or restore normal glucose metabolism. Increased reliance on fat metabolism promotes ketone body production, increasing the risk of diabetic ketoacidosis, and may also increase the risk of cardiovascular disease. Therefore, it is not an effective primary treatment.***
- ***The most appropriate management is insulin therapy combined with a balanced diet, in which carbohydrate intake is matched with insulin dosage. This maintains blood glucose within the normal range while minimising both hyperglycaemia and hypoglycaemia.***

15. A booming cosmetic clinic in Kampala introduced a fast-acting **"organic" skin-lightening serum**. After two months of use, patients returned to the clinic complaining that their skin had become fragile and translucent, with superficial blood vessels becoming more visible. Their skin was dehydrated, severely blistered, and unable to heal from minor scratches or wounds.

A dermatologist teamed up with a molecular biologist to investigate the effect of the serum on the skin epidermal tissue. The results of the investigations were summarised in the table below.

To manage the severe side effects in the affected patients and keep the serum on the market, the consulting doctors proposed prescribing a topical barrier ointment highly concentrated with synthetic cholesterol, to be applied twice daily, or performing a mechanical dermabrasion procedure to remove the blistered epidermal layers. However, the dermatologist suggested that a synthetic growth factor drug would be more beneficial because it would strongly stimulate the growth of skin basal cells.

A table showing epidermal tissue analysis results

Parameter	Healthy Skin before treatment	Affected Skin after treatment

DNA polymerase enzyme activity (units)	45	4
Cellular cholesterol-to-phospholipid ratio	1:1	1:4
Intracellular water volume	Normal	Low/decreased
Proportion of cells stalled in the S-phase of the cell cycle (%)	<2	68

Task:

(a) Explain how the organic serum caused the observed symptoms in affected patients.

- ***The serum reduced DNA polymerase activity from 45 to 4 units in epidermal cells, limiting DNA replication during the S phase of the cell cycle. This caused 68% of cells to stall in S phase compared with less than 2% in healthy skin, reducing the formation of new epidermal cells.***
- ***Consequently, the epidermis became thin and fragile, and the weakened epidermal barrier allowed superficial blood vessels to become more visible, giving the skin a translucent appearance. Reduced epidermal cell replacement also slowed repair of minor scratches and wounds.***
- ***The serum lowered the cholesterol-to-phospholipid ratio from 1:1 to 1:4. Reduced cholesterol made epidermal cell membranes less stable and more permeable, increasing cell damage and weakening the skin barrier. This contributed to severe blister formation.***
- ***The serum reduced intracellular water volume from normal to low levels, causing dehydration of epidermal cells. Dehydrated cells lose their normal structure and function, making the skin dry, fragile, and less capable of protecting underlying tissues.***

(b) Evaluate the biological safety of each of the proposed interventions.

- ***Synthetic cholesterol barrier ointment** may improve skin protection by increasing cholesterol availability in the epidermal barrier, restoring membrane stability and reducing water loss from skin cells. However, it does not restore DNA polymerase activity or stimulate production of new epidermal cells, so it may not fully correct the impaired wound healing and thin epidermis. It is therefore useful as supportive treatment but not a complete solution.*
- ***Mechanical dermabrasion** may remove damaged blistered epidermal layers, allowing healthier tissue beneath to be exposed. However, because the epidermis has reduced cell renewal capacity, removal of the protective layer may increase the risk of infection, dehydration, and delayed healing. Therefore, it may worsen the condition and is biologically unsafe in severely affected patients.*
- ***Synthetic growth factor drug** can stimulate basal cells to divide and replace damaged epidermal cells, addressing the reduced skin regeneration caused by low DNA polymerase activity and S-phase arrest. However, excessive stimulation of cell division may cause uncontrolled cell growth or abnormal tissue formation if not carefully regulated. Therefore, it may be the most beneficial option when used at controlled doses and under medical supervision.*
- ***Therefore, the growth factor treatment is the most likely to restore epidermal repair, while the cholesterol ointment can provide supportive protection. Mechanical dermabrasion should be avoided or used with caution because it removes an already weakened protective barrier.***

16. During a national marathon training camp in Kapchorwa, athletes were given a new energy supplement called **"Mega-Boost"**, which was claimed to increase cellular energy production instantly. However, several athletes who consumed the supplement developed severe muscle fatigue and eventually collapsed during long-distance races despite having normal blood glucose concentrations.
Sports physiologists investigated the effect of the supplement on aerobic respiration in muscle cells. Laboratory analysis showed that the supplement supplied ATP directly to muscle cells while interfering with oxidative phosphorylation, causing NADH to accumulate within the mitochondria. Electron microscopy further showed that prolonged use of the supplement caused swelling of the mitochondrial cristae in muscle fibres. The results obtained are shown in the table below.

A table showing effects of "Mega-Boost" on muscle cells

Parameter	**Normal athlete**	**Athlete using "Mega-Boost"**
ATP concentration in muscle cells (mmol kg^{-1} muscle)	5.2	18.6
NADH concentration	Normal	Very high
Pyruvate concentration in cytoplasm (mmol L^{-1})	3.3	1.2
Lactic acid concentration in muscle cells (mmol L^{-1})	1.5	14.2
Rate of carbohydrate breakdown	High	Very low
Rate of lipid breakdown	Moderate	Very high

Following the collapses, the medical team attempted the following interventions:

- Administering **100% oxygen** during recovery.
- Giving the athletes a **high-glucose syrup** immediately after they collapsed.
- Administering a drug that **competitively inhibits fatty acid oxidation (β-oxidation)** before the race.

Task:
(a) Analyse the findings to explain how the **Mega-Boost** supplement affected the cellular and biochemical processes in muscle cells, and how these changes led to the athletes' collapse.

- ***The supplement increased ATP concentration in muscle cells from 5.2 to 18.6 mmol kg^{-1} of muscle tissue by directly supplying ATP. The high ATP concentration exerted feedback inhibition on phosphofructokinase, reducing the rate of glycolysis. Consequently, carbohydrate breakdown became very low, and less pyruvate was produced, causing cytoplasmic pyruvate concentration to decrease from 3.3 to 1.2 mmol L^{-1}.***
- ***The supplement interfered with oxidative phosphorylation by preventing NADH from effectively donating electrons to the electron transport chain. As a result, NADH accumulated***

within the mitochondria, reducing the regeneration of NAD^+ required for continued aerobic respiration.

- ***The shortage of NAD^+ limited the complete aerobic oxidation of pyruvate because pyruvate could no longer be efficiently oxidized in the mitochondria. Consequently, muscle cells increasingly relied on anaerobic respiration, in which pyruvate was reduced to lactic acid to regenerate NAD^+ and allow glycolysis to continue. This caused lactic acid concentration to increase markedly from 1.5 to 14.2 mmol L^{-1}. The accumulated lactic acid lowered muscle pH, impaired enzyme activity and reduced muscle contraction efficiency, leading to severe muscle fatigue.***
- ***Because carbohydrate metabolism was greatly reduced, muscle cells increasingly relied on lipids as respiratory substrates, resulting in a very high rate of lipid breakdown. However, β-oxidation also produces NADH, which could not be efficiently reoxidized because oxidative phosphorylation was impaired. Therefore, increased lipid breakdown could not provide a sustainable supply of ATP during prolonged exercise.***
- ***Prolonged use of the supplement caused swelling of the mitochondrial cristae, reducing the surface area available for the electron transport chain and ATP synthase. This further impaired oxidative phosphorylation and reduced the muscles' ability to regenerate ATP aerobically. Although ATP concentration was elevated to 18.6 mmol kg^{-1} of muscle tissue due to the direct supply of ATP by the supplement, this ATP could not be continuously replenished during prolonged exercise because mitochondrial ATP production was impaired. As the available ATP was consumed, muscle cells experienced an energy deficit, contributing to severe fatigue and eventual collapse.***
- ***Although the athletes had normal blood glucose concentrations, the problem was not glucose availability but the inability of muscle cells to efficiently convert respiratory substrates into usable ATP due to disrupted mitochondrial function.***
- ***Therefore, the combined effects of impaired oxidative phosphorylation, NADH accumulation, increased dependence on anaerobic respiration, lactic acid accumulation, and mitochondrial damage caused collapse during long-distance races.***

(b) Evaluate the effectiveness of each intervention used by the medical team.

- ***Administering 100% oxygen during recovery increases oxygen availability to muscle cells, which may support aerobic respiration when mitochondrial function begins to recover. However, the main problem is impaired oxidative phosphorylation rather than insufficient oxygen supply. Therefore, oxygen therapy may assist recovery but cannot directly correct the disruption of the electron transport chain caused by the supplement.***
- ***Giving high-glucose syrup immediately after collapse provides additional glucose for energy production and may help restore glycogen reserves. However, glycolysis is already reduced due to feedback inhibition by high ATP concentration, and oxidative phosphorylation remains impaired. Therefore, the additional glucose cannot be efficiently converted into ATP, making this intervention less effective for immediate recovery.***
- ***Administering a drug that competitively inhibits fatty acid oxidation (β-oxidation) before the race may reduce excessive production of NADH from lipid breakdown and limit further NADH accumulation in mitochondria. However, fatty acid oxidation is an important energy source during prolonged exercise, and reducing it would further limit energy availability if carbohydrate metabolism remains suppressed. Therefore, it is unlikely to be an effective primary solution.***

- ***In a nutshell, the most effective intervention is to stop the use of Mega-Boost, allowing mitochondrial structure and normal oxidative phosphorylation to recover. Supportive measures such as oxygen therapy, adequate hydration, rest, and gradual recovery should accompany withdrawal of the supplement to restore normal muscle function.***

BIOLOGY PRACTICAL (P530/2)

Biology is a practical science that seeks to explain living systems through observation, experimentation, and evidence-based reasoning. The A'Level Biology Practical examination is therefore designed to assess not only a learner's understanding of biological concepts, but also their ability to apply scientific knowledge and skills to unfamiliar situations.

The practical paper emphasizes critical thinking, problem-solving, scientific investigation, and the interpretation of biological evidence. Learners are expected to demonstrate competence in making observations, taking measurements, drawing biological specimens, planning and conducting investigations, analysing data, drawing conclusions, and making justified recommendations. Furthermore, candidates should be able to relate structural and behavioural adaptations of organisms to their survival and success in different environments.

Practical paper structure

The A'Level Biology Practical paper consists of two compulsory, scenario-based items and is completed within a period of 3 hours. Items may be drawn from any of the four Biology constructs and their respective assessment objectives.

- One item assesses observational investigation skills, requiring candidates to obtain and interpret data from specimens and relate structural and behavioural mechanisms to the survival advantage of organisms.
- The second item assesses scientific investigation skills. Candidates are expected to:
    - Identify and analyse challenges presented in a scenario.
    - Plan and conduct investigations.
    - Collect, present, and interpret data.
    - Draw evidence-based conclusions.
    - Make justified recommendations based on their findings.

The practical paper is designed to assess a learner's ability to think and work like a scientist by applying biological principles to authentic and meaningful contexts.

For example, in an item involving the dissection of a mammal (rat), a candidate is expected to demonstrate a high level of practical competence. To obtain the highest achievement level, the candidate should:

i. Produce a clear and logical procedure for dissection by:

- Correctly orienting the specimen.

- Opening the body cavity to expose the internal structures.
- Systematically displacing internal organs to display the required structures.

ii. Produce accurate and well-labelled biological drawings of the observed structures.
iii. Make systematic observations supported by accurate measurements of both external and internal structures.

These expectations highlight the importance of precision, organization, and scientific communication in Biology practical work.

This section of the book presents practical items with model responses and self-test activities to help learners develop the confidence, precision, and scientific inquiry skills required for excellence in A'Level Biology and for further studies in the biological sciences.

PAPER 2 (P530/2) ITEMS WITH RESPONSES

The practical items have been classified into two categories: **Observational Investigation Skills** and **Scientific Investigation**. In the first category, particular emphasis is placed on practical items with their corresponding responses about rat dissection and microscopy, presented in this book.

1.0. OBSERVATIONAL INVESTIGATION SKILLS

1.1. Mammal A feeds on plant material W. The tuber is digested in the gut, and the products of digestion are absorbed into the body to benefit the animal. The gut is structurally adapted to perform its roles. The products of digestion of the plant material are absorbed into the bloodstream at different regions of the gut. The cells of the plant material are adapted to enable the animal to benefit after digestion and absorption of nutrients into the body.

Mammal B also feeds on plant material W and digests it in its gut before absorbing the products of digestion into the body. Animal B has the following hypothetical data for some parts of its gut.

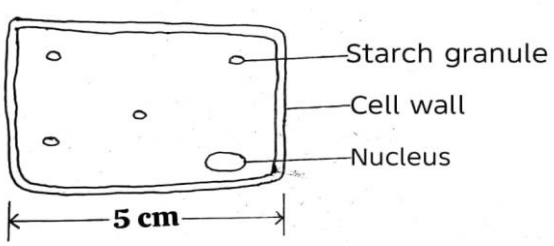
Organ	Length (mm)
Stomach	60
Duodenum	30
Ileum	850

Task:

(a) Using a microscope under medium power, observe a thin slide prepared from plant material W. Record the number of cells visible in the field of view and calculate the linear magnification of one clearly visible cell. Hence, state how the cell is adapted to benefit the animal.

Number of cells in field of view = cells in row x cells in column
= 15 x 10 = 150 cells
Size of field of view = 2 mm = (2x1000) µm = 2,000 µm

A drawing of one clearly visible cell of plant material W under medium power of a light microscope



Features of plant material W observed under a microscope:

- ***Many parenchyma cells***
- ***Many starch granules (amyloplasts)***
- ***Polygonal cells***

Calculation of the actual cell size (medium power x100):

10 cells occupy 2 mm in length

1 cell = $\frac{2}{10}$ = 0.2 mm

Magnification = $\frac{Image\ size}{Actual\ size}$

$$100 = \frac{0.2}{Actual\ size}$$

Actual size = $\frac{0.2}{100} = 0.002\ mm$

0.002mm = (1000x0.002) µm = 2 µm

Calculation of the linear magnification

Measured length of the drawn cell/ image = 5 cm

5 cm = 50 mm = 50,000 µm

Linear magnification = $\frac{length of\ drawn\ cell}{length\ of\ actual\ cell} = \frac{50{,}000}{2} = \times 25{,}000$

Therefore, the linear magnification is ×25,000

Adaptations of cells and tissues of plant material W

- ***Many parenchyma cells provide a large number of storage cells for accumulating large quantities of starch, making the tuber a rich food source for the animal.***
- ***Numerous starch granules allow large quantities of starch to be stored within the parenchyma cells, providing abundant food for the animal after digestion.***
- ***Polygonal-shaped parenchyma cells fit closely together, allowing efficient packing of storage cells within the tuber tissue.***

(b) Dissect Mammal A to investigate the gut. Outline the steps taken during the dissection. Record the lengths of similar organs in the gut of Mammal A and those of Mammal B in the table provided, in millimetres. Provide a vivid explanation of the

significance of the results in the table, and use the data to compare the efficiency of each animal's gut. Make a well-labelled drawing of the blood vessels supplying the measured structures.

Organ measurements for animal A compared to those of animal B

Organ	Animal B	Animal A
Stomach length (mm)	60	**120**
Duodenum length (mm)	30	**80**
Ileum length (mm)	850	**1030**

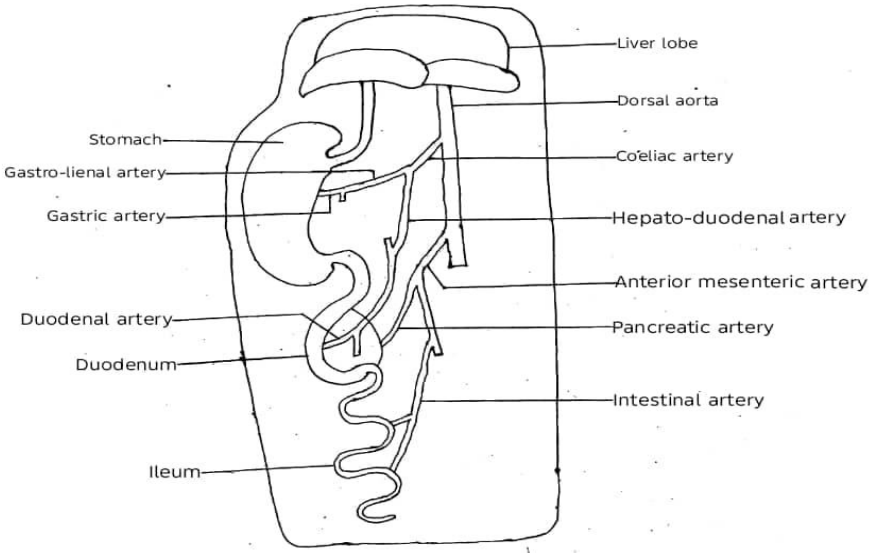
Explanation

- ***Larger and longer stomach, increasing the capacity for storage of more food than Animal B.***
- ***Longer duodenum and ileum in Animal A than Animal B, providing more space for digestion and absorption of nutrients.***
- ***Animal A is larger than Animal B, hence requires longer organs to accommodate its greater food intake and meet its nutritional requirements.***

Procedure for dissection of specimen A

1. ***Place Mammal A ventral side uppermost on the dissecting board and pin it through the feet.***
2. ***Lift the skin at the mid-ventral line and make an incision along this line, extending from the lower jaw to the genital region.***
3. ***Loosen the skin from the body wall and pin it back on both sides of the animal.***
4. ***Make a mid-ventral incision through the body wall from the diaphragm to the pelvic girdle, taking care not to damage the underlying organs.***
5. ***Pin back the separated body wall to expose the abdominal viscera clearly.***
6. ***Displace the liver lobes anteriorly to expose the stomach.***
7. ***Displace the caecum and colon to the lower left to expose the stomach, duodenum and ileum clearly on the right side of the animal.***

A drawing of blood vessels supplying blood to the stomach, duodenum and ileum of animal A with structures displayed to the right of the animal

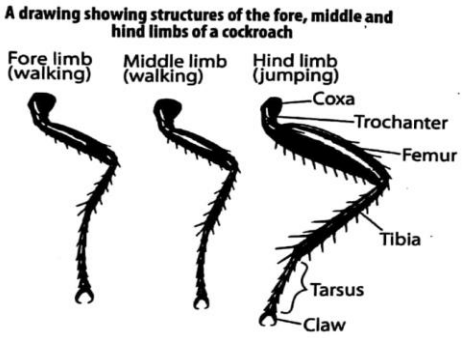


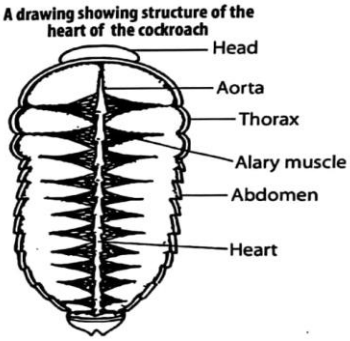
X1

1.2. At the end of the first term, Peter left his leather suitcase in the dormitory and went home for the holidays. When he returned for the second term, he opened the suitcase and a rat jumped out. The suitcase had been torn, with cockroaches hiding among his books and potato crisps partly eaten. Disturbed by the situation, Peter reported the matter to the warden, who referred it to the headteacher.
The headteacher called a pest control expert, who fumigated the dormitory and set traps that caught several rats. To help the students understand the biology of the infestation, the headteacher invited the biology teacher to explain the situation during assembly. Using a trapped rat and charts showing cockroach mouthparts and limbs, the teacher explained how both animals feed on potato crisps because the tissues of the potato tuber store food that can be observed under a microscope. He further explained that the rat was able to escape quickly because of the size and efficiency of its locomotory structures. To illustrate his explanation, he displayed drawings of cockroach limbs and the heart, along with a table showing limb measurements, as shown below.

Table showing relative length and thickness of limbs

Limb type	Relative length (cm)	Relative thickness (mm)
Fore limb	1.5	1.5
Middle limb	2.0	2.0
Hind limb	2.8	3.0





The headteacher and students were not convinced by the explanation and required a detailed biological interpretation to help them understand how rats and cockroaches had infested the dormitory and how the problem could be controlled.
You are provided with a rat (specimen X) a potato tuber (specimen Y), and other apparatus to help the headteacher and students interpret the biology teacher's observations.

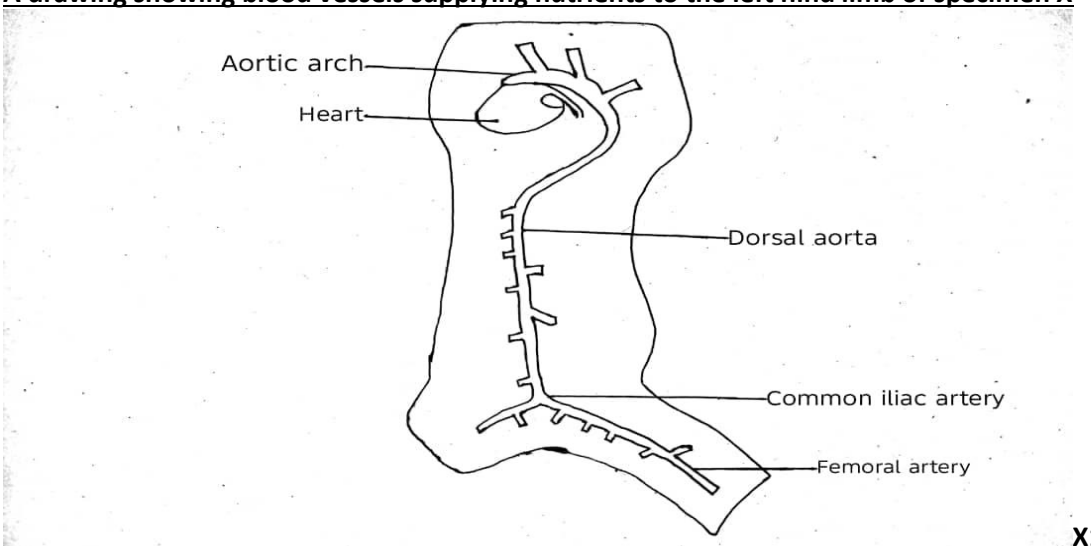
Task:
(a) Observe the locomotory structures of specimen X and dissect it to display the transport structures supplying the limbs. Draw and label these structures and explain the teacher's findings.

Procedure for dissection:

1. ***Specimen E was placed ventral side uppermost on the dissecting board and pinned through the feet.***
2. ***The skin was lifted at the mid-ventral line, and an incision was made along this line from the lower jaw to the genital region.***
3. ***The skin was loosened from the body wall and pinned back on both sides of the specimen.***
4. ***A mid-ventral incision was made through the abdomen from the diaphragm to the pelvic girdle.***
5. ***The separated body wall was pinned back to expose the abdominal viscera clearly.***
6. ***The liver lobes were displaced anteriorly to expose the stomach and oesophagus.***
7. ***The hepatic portal vein was traced from the liver, ligatured, and cut at the distal end of the ligature.***
8. ***The stomach and ileum were gently displaced to the right to expose the coeliac and anterior mesenteric arteries, which were ligatured and cut at the distal end of each ligature.***
9. ***The oesophagus and rectum were gently cut to allow removal of the gut from the abdominal region, taking care not to damage the posterior mesenteric artery.***
10. ***The muscles of the pelvic girdle were cut to expose the pubic symphysis without damaging the arteries and veins supplying the hind limbs. The legs were then widened and re-pinned.***
11. ***The xiphoid cartilage was held with forceps and raised. The falciform ligament was cut with scissors to allow the liver to fall further away from the diaphragm. The diaphragm was then cut away to expose the path of the dorsal aorta.***
12. ***The xiphoid cartilage was held firmly and cut with scissors through the chest cavity posteriorly up to the clavicle bones.***

13. ***Parts of the rib cage were cut away on both sides to expose the arches, blood vessels and heart. Connective tissue, the thymus gland and fat tissue were removed from around the heart.***
14. ***The blood vessels supplying the limbs were traced.***

A drawing showing blood vessels supplying nutrients to the left hind limb of specimen X



X1

A table showing the relative length and thickness of limbs of specimen X (rat)

Limb type	Relative length (cm)	Relative thickness (mm)
Fore limb	5.6	40
Hind limb	8.0	65
Tail	13.4	12

In comparison, the rat is a more adapted terrestrial organism than the cockroach because:

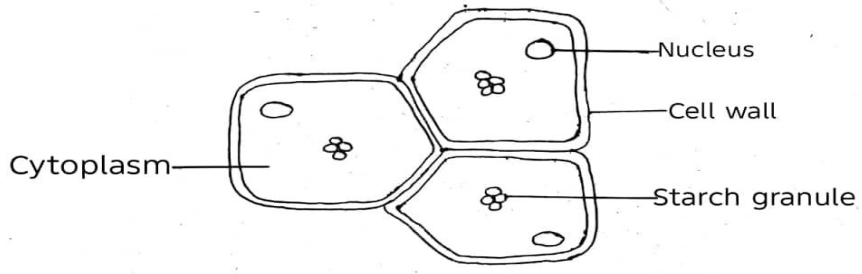
- ***The hind limbs of the rat are four times longer than those of the cockroach, allowing faster locomotion.***
- ***The hind limbs are 25 times thicker and more muscular than those of the cockroach, providing a stronger propulsion force during locomotion.***
- ***Possession of a long tail, which is almost equal in length to the rest of the body, provides balance during fast locomotion.***
- ***Possession of an efficient circulatory system for the rapid transportation of nutrients and respiratory gases to the limbs, leading to a high metabolic output because a lot of energy is generated.***
- ***Possession of an endoskeleton with joints provides stronger attachment of thick muscles and support for body weight during locomotion.***

(b) (i) Using a microscope, draw and label three representative structures from the stem of specimen Y to show why the crisps were described as dietary food for rats and cockroaches.

Procedure for microscopy:

- ***Using a sharp blade, a thin section of the tissue is cut and placed on the glass slide.***
- ***Three drops of iodine solution are added to the tissue, and it is allowed to stain for 5 minutes.***
- ***Using filter paper, excess iodine solution is removed, and the tissue is covered with a coverslip.***
- ***The slide is mounted on the stage of the microscope and observed under low/medium power of the microscope.***

A drawing showing the structure of three representative cells of specimen Y under low/medium power of a light microscope



(ii) From your drawing, explain why the observed structures are considered dietary food for these pests.

The tissues possess:

- ***Numerous parenchyma cells to increase surface area for storage of starch granules.***
- ***Closely packed starch granules for increased food storage in the cells.***

2.0. SCIENTIFIC INVESTIGATION

2.1. Joan is piloting a project in which she intends to generate and use the gas for study in a school laboratory. She produces the gas by the catalytic breakdown of solution **P** at a constant temperature. She was advised to use an enzyme obtained from plant extracts. Joan obtains her enzyme from extract **C**, which is prepared by mixing 4 g of peeled and crushed Irish potato with 50 cm^3 of distilled water. The mixture is stirred and decanted.

However, she first needs to find out whether the extracts on the market labelled **A** and **B** could produce more gas per unit time than her local extract **C**.
To further increase the rate of gas production, Joan was advised by experts to add an enzyme activator to her setup. Currently, there are two types of enzyme activators on the market, activators X and Y, and she needs to be guided on which one increases gas production in her case.

You are provided with samples of solution P, enzyme activators **X** and **Y**, enzyme extracts **A** and **B**, and specimen **C**, from which Joan prepared her enzyme extract **C**. Joan needs guidance on how to select the appropriate materials for her gas production.

Task:
Design and carry out an investigation, then use your findings to give appropriate advice to Joan in the form of a report.
Your response should be in the form of a scientific report.

Response:

Aim: To determine which combination of enzyme extract A, B or C and activator X or Y maximises gas production from the breakdown of solution P and recommend it to Joan.

Hypothesis: *Extract A would produce the highest rate of gas production when combined with activator X because it contains the highest enzyme concentration and activator X provides more favourable conditions for enzyme activity.*

Variables:
Independent variables:
- *Concentration of enzyme extracts A, B and C.*
- *Type of activator, X or Y.*

Dependent variable:
- *Volume or rate of gas produced.*

Controlled variables:
- *Volume of each solution or extract used.*
- *Temperature of the reaction.*
- *Duration of the experiment.*

Risks and Mitigations:
- ***Risk 1:*** *Reagents may cause skin burns on contact due to their corrosive nature.*
- ***Mitigation:*** *The reagents were handled carefully, and protective gloves were worn throughout the experiment.*
- ***Risk 2:*** *Careless handling of knives may cause injury.*
- ***Mitigation:*** *All cutting instruments were handled with care.*

Procedure:
Effect of enzyme concentration using extracts A, B and C on gas production
1. *Three test tubes were labelled A, B and C.*
2. *2 cm^3 of the respective extract A, B or C was added to each labelled test tube.*
3. *2 cm^3 of solution P was added to test tube A, and the rate of bubbling was recorded.*
4. *Step 3 was repeated for test tubes B and C, and the observations were recorded.*

Effect of activators X and Y on enzyme activity
1. *Two test tubes were labelled X and Y.*

2. *2 cm³ of extract A was added to each test tube.*
3. *Two drops of activator X were added to test tube X, while two drops of activator Y were added to test tube Y.*
4. *2 cm³ of solution P was added to each test tube, and the rate of bubbling was recorded.*

Results:

Solution mixture	Observation
P + A	*Rapid effervescence*
P + B	*Moderate effervescence*
P + C	*Slow effervescence*
P + A + X	*Rapid effervescence*
P + A + Y	*Slow effervescence*

Explanation:

- *Increasing enzyme concentration increases the frequency of enzyme-substrate collisions, leading to the formation of more enzyme-substrate complexes and consequently increasing the rate of gas production.*
- *Activator **X** enhanced the rate of reaction by modifying the enzyme or substrate to facilitate the formation of enzyme-substrate complexes.*
- *Activator **Y** inhibited enzyme activity by providing unfavourable conditions for the reaction, thereby reducing the rate of gas production.*

Conclusion:

- *The results confirmed that increasing enzyme concentration and using activator **X** increased the rate of gas production, consistent with the stated hypothesis.*

Recommendation:

- *Joan should use extract **A**, which produced the highest rate of gas production, combined with activator **X** to achieve the greatest rate of gas production in her project.*

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